bsnr — Bioacoustic SNR Estimation: Gallery of Examples

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This gallery demonstrates the main features of the bsnr package for estimating signal-to-noise ratio in bioacoustic recordings.

All examples use the same synthetic fixtures as the bsnr test suite (`createTestFixture`, `createCalibratedTestFixture`, `makeSRWUpcall`, `makeClickAudio`), ensuring consistency between documentation and tests.

To publish this gallery as HTML:

```
cd C:\analysis\bsnr\examples
publish('bsnr_gallery.m', 'format', 'html', 'outputDir', 'html')
```

Requirements: MATLAB Signal Processing Toolbox. See README.md for the full dependency list and installation instructions.

Setup

Add bsnr source, examples, and tests directories to the path.

```
thisDir   = fileparts(mfilename('fullpath'));
sourceDir = fileparts(thisDir);
testsDir  = fullfile(sourceDir, 'tests');
addpath(sourceDir, '-begin');
addpath(thisDir,   '-begin');
addpath(testsDir);

% Add external dependencies if present
for dep = {'longTermRecorders','annotatedLibrary','bsmTools','soundFolder'}
    candidate = fullfile('C:\analysis', dep{1});
    if exist(candidate,'dir'), addpath(candidate); end
end

% Shared spectrogram display parameters (2000 Hz recordings)
sp.pre        = 1;
```

```
sp.post       = 1;
sp.yLims      = [0 500];
sp.freq       = [150 250];
sp.noiseDelay = 0;
sp.win        = floor(2000 / 4);
sp.overlap    = floor(sp.win * 0.75);

close all;
```

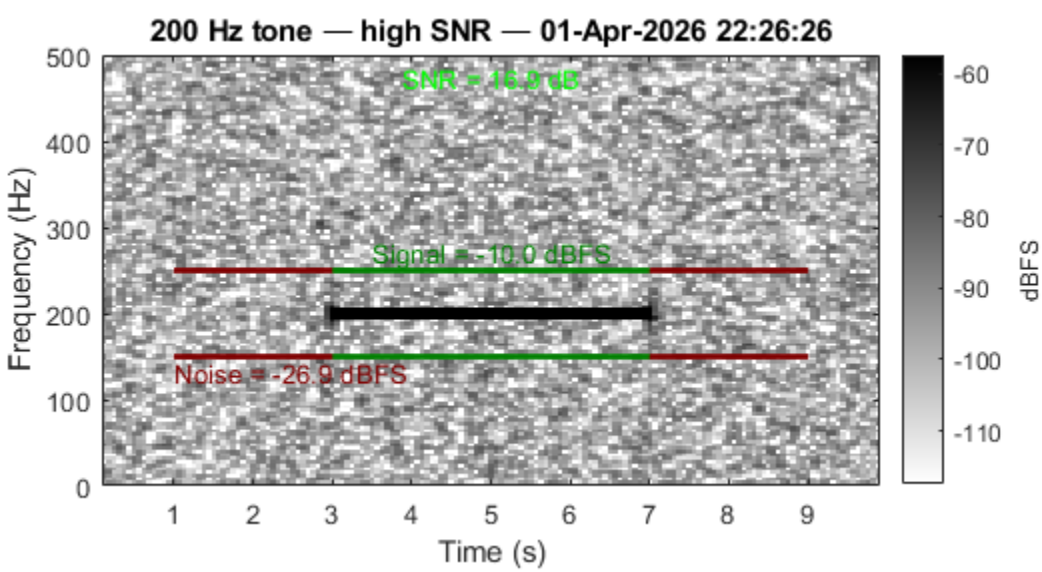
1. Basic SNR Estimation

`snrEstimate` measures the signal-to-noise ratio of a bioacoustic detection directly from a WAV file. The detection is described by a struct with fields `soundFolder`, `t0`, `tEnd`, `freq`, and `channel`. The noise window is positioned automatically adjacent to the detection.

Here we generate a synthetic WAV using `createTestFixture` — a 200 Hz tone at approximately 20 dB in-band SNR — and run the spectrogram method.

```
[annot1, cleanup1] = createTestFixture( ...
    'signalRMS', 1.0, 'noiseRMS', 0.1, 'toneFreqHz', 200, ...
    'freq', [150 250], 'durationSec', 4, ...
    'classification', '200 Hz tone — high SNR');

p1 = struct('snrType', 'spectrogram', 'showClips', true, ...
    'pauseAfterPlot', false, 'spectroParams', sp);

figure('Position', [100 100 560 280], 'Name', 'Basic SNR estimation');
[snr1, rmsS1, rmsN1] = snrEstimate(annot1, p1);
fprintf('SNR = %.1f dB  |  signal = %.3g  |  noise = %.3g\n', snr1, rmsS1,
 rmsN1);
cleanup1();
```

```
Reading audio metadata for file 1/1: C:\Users\brian_mil\AppData\Local\Temp
\annotSNR_test_20260401_222624_190\2026-04-01_22-26-24.wav
SNR = 16.9 dB  |  signal = 0.0999  |  noise = 0.00203
```



2. Frequency-Modulated Call — SRW Upcall

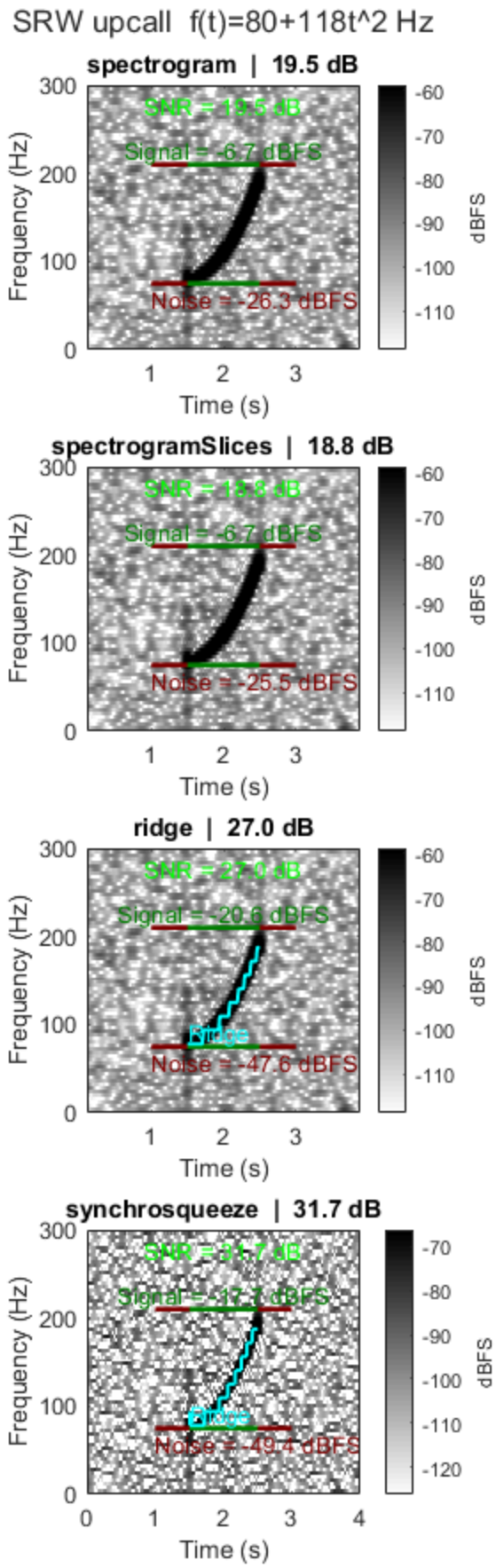
Ridge tracking and synchrosqueezing are specifically designed for frequency-modulated tonal calls. Here we use a synthetic Southern Right Whale upcall generated by `makeSRWUpcall`, following the quadratic frequency law $f(t) = 80 + 118 \cdot t^2$ Hz (sweeping $_{80 \to 198}$ Hz over 1 s).

The cyan overlay on the ridge and synchrosqueeze tiles shows the tracked instantaneous frequency contour.

```
srwRate     = 1000;
srwFreq     = [75 210];
bufferSec   = 2;
srwNoiseRMS = 0.1;

[srwSig, ~] = makeSRWUpcall(srwRate, srwNoiseRMS);
rng(7);
bufSamps    = round(bufferSec * srwRate);
srwWideRMS  = srwNoiseRMS * sqrt(srwRate/2 / diff(srwFreq));
fullAudio   = [srwWideRMS*randn(bufSamps,1); srwSig;
 srwWideRMS*randn(bufSamps,1)];
fullAudio   = fullAudio * (0.9/max(abs(fullAudio)));

srwDir = fullfile(tempdir, sprintf('bsnr_gallery_srw_%s',
 datestr(now,'yyyymmdd_HHMMSS')));
mkdir(srwDir);
srwStart = floor(now()*86400)/86400;
audiowrite(fullfile(srwDir, [datestr(srwStart,'yyyy-mm-dd_HH-MM-
SS') '.wav']), ...
    fullAudio, srwRate);
srwCleanup = @() rmdir(srwDir,'s');

srwAnnot.soundFolder    = srwDir;
srwAnnot.t0             = srwStart + bufferSec/86400;
srwAnnot.tEnd           = srwStart + (bufferSec + 1.0)/86400;
srwAnnot.duration       = 1.0;
srwAnnot.freq           = srwFreq;
srwAnnot.channel        = 1;
srwAnnot.classification = 'SRW upcall f(t)=80+118t^2 Hz';

srwSP         = sp;
srwSP.yLims   = [0 300];
srwSP.freq    = srwFreq;
srwSP.win     = floor(srwRate/4);
srwSP.overlap = floor(srwSP.win * 0.9);

fig4 = figure('Position', [190 190 300 896], 'Name', 'SRW upcall — FM
 methods');
tlo4 = tiledlayout(fig4, 4, 1, 'TileSpacing','compact','Padding','compact');
title(tlo4, 'SRW upcall  f(t)=80+118t^2 Hz', 'interpreter','none');

for snrType = {'spectrogram','spectrogramSlices','ridge','synchrosqueeze'}
    nexttile(tlo4);
    p = struct('snrType', snrType{1}, 'showClips', true, ...
        'pauseAfterPlot', false, 'spectroParams', srwSP);
```

```
    snr = snrEstimate(srwAnnot, p);
    title(gca, sprintf('%s  |  %.1f dB', snrType{1},
 snr), 'interpreter','none');
end
srwCleanup();
```

Reading audio metadata for file 1/1: C:\Users\brian_mil\AppData\Local\Temp\bsnr_gallery_srw_20260401_222624\2026-04-01_22-26-24.wav


SRW upcall f(t)=80+118t^2 Hz
spectrogram | 19.5 dB
SNR = 19.5 dB
Signal = -6.7 dBFS
Noise = -26.3 dBFS
spectrogramSlices | 18.8 dB
SNR = 18.8 dB
Signal = -6.7 dBFS
Noise = -25.5 dBFS
ridge | 27.0 dB
SNR = 27.0 dB
Signal = -20.6 dBFS
Ridge
Noise = -47.6 dBFS
synchrosqueeze | 31.7 dB
SNR = 31.7 dB
Signal = -17.7 dBFS
Ridge
Noise = -49.4 dBFS
Frequency (Hz)
Time (s)
dBFS

3. Calibrated Acoustic Levels

When instrument metadata is supplied via `params.metadata`, the spectrogram colour scale shifts from normalised WAV units to calibrated acoustic units (dB re 1 µPa²/Hz), and the result table gains two columns:

- `signalBandLevel_dBuPa` — band-integrated signal level (dB re 1 µPa)
- `noiseBandLevel_dBuPa` — band-integrated noise level (dB re 1 µPa)

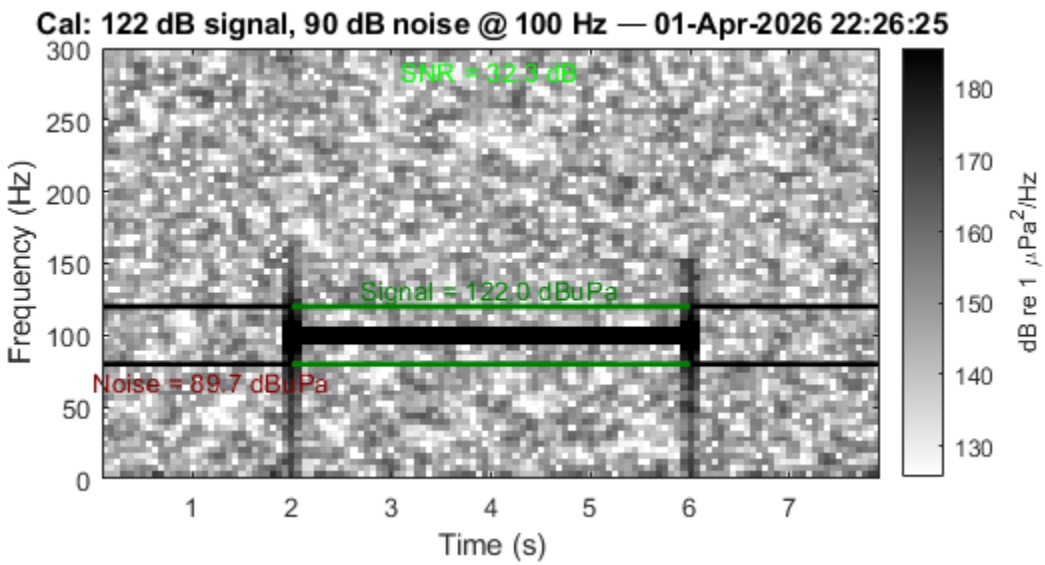
These are equivalent to `bandpower(psdCal, f, freq, 'psd')` from a fully calibrated PSD — correct for both tonal and broadband signals.

The instrument model is `simpleFlatMetadata`: flat 20 dB gain with realistic AC coupling (~5 Hz corner) and anti-aliasing (~5 kHz corner). Signal: 100 Hz tone at 122 dB re 1 µPa. Noise: 90 dB re 1 µPa (32 dB SNR).

```
% createCalibratedTestFixture generates a WAV with a tone at a known
% acoustic level using the instrument metadata chain. The spectrogram
% colour scale shows dB re 1 µPa²/Hz and signal/noise labels show dBuPa.
% Signal: 122 dB re 1 µPa at 100 Hz. Noise: 90 dB re 1 µPa. SNR: 32 dB.
[annot5, meta5, cleanup5] = createCalibratedTestFixture( ...
    'signalLeveldB', 122, 'noiseLeveldB', 90, ...
    'toneFreqHz', 100, 'freq', [80 120], 'durationSec', 4);

sp5            = sp;
sp5.freq       = [80 120];
sp5.yLims      = [0 300];
sp5.win        = floor(meta5.sampleRate / 4);
sp5.overlap    = floor(sp5.win * 0.75);
% bufferSec = 0.5*durationSec = 2s. Noise window is placed AFTER the
% detection (noiseDelay = durationSec) so it falls within the file.
% pre/post = 0 avoids requesting audio outside the file boundaries.
sp5.pre        = 0;
sp5.post       = 0;
sp5.noiseDelay = annot5.duration;   % noise window after detection

figure('Position', [220 220 560 280], 'Name', 'Calibrated spectrogram');
try
    p5 = struct('snrType', 'spectrogram', 'showClips', true, ...
        'pauseAfterPlot', false, 'metadata', meta5, 'spectroParams', sp5, ...
        'freq', [80 120]);
    [snr5, rmsS5, rmsN5] = snrEstimate(annot5, p5);
    fprintf('SNR = %.1f dB | signal = %.1f dBuPa | noise = %.1f dBuPa\n', ...
        snr5, 10*log10(rmsS5), 10*log10(rmsN5));
catch err5
    fprintf('Calibrated section error: %s\n', err5.message);
    for ii = 1:numel(err5.stack)
        fprintf('  at %s (line %d)\n', err5.stack(ii).name,
 err5.stack(ii).line);
    end
end
cleanup5();
```

```
Reading audio metadata for file 1/1: C:\Users\brian_mil\AppData\Local\Temp
\annotSNR_cal_20260401_222625_075\2026-04-01_22-26-25.wav
```

```
SNR = 32.3 dB | signal = 122.0 dBuPa | noise = 89.7 dBuPa
```



4. Click Contamination and Removal

Impulsive noise — echosounder pings, mechanical transients, or flow noise spikes — inflates power-based SNR estimates, particularly for `snrTimeDomain` and `snrRidge` which measure instantaneous amplitude.

`removeClicks` applies the PAMGuard soft amplitude gate:

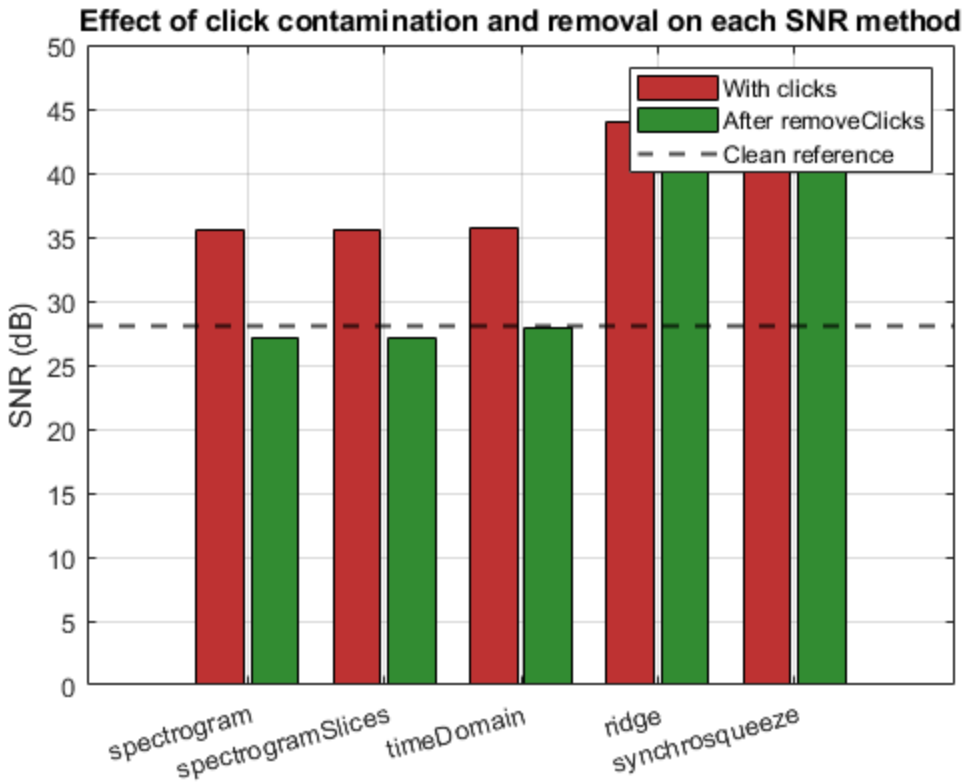
```
w(x) = 1 / (1 + (x/threshold)^power)
```

with `threshold = 3 * std(audio)` and `power = 1000` (effectively a hard gate). The left bars show SNR with clicks present; the right bars show SNR after removal. The dashed line marks the clean reference SNR.

Clicks are 5 ms sine bursts at the tone frequency, repeating every 1 s at 50× the signal amplitude — well above the detection threshold.

```
[sigWithClicks, sigClean, noiseAudio] = makeClickAudio( ...
    2000, 10, 200, 1.0, 0.1, 1.0, 50.0);

nfft     = 2^nextpow2(floor(10/30/0.75*2000));
nOverlap = floor(nfft*0.75);
sigCleaned = removeClicks(sigWithClicks, 3, 1000);

% Reference SNR from clean signal
[rSref, rNref] = snrTimeDomain(sigClean, noiseAudio, [150 250], 2000);
refSNR = 10*log10(rSref/rNref);

clickMethods =
 {'spectrogram','spectrogramSlices','timeDomain','ridge','synchrosqueeze'};
nM = numel(clickMethods);
snrContam  = nan(nM,1);
snrCleaned = nan(nM,1);
```

```
for k = 1:nM
    m = clickMethods{k};
    switch m
        case 'spectrogram'
            [rS,rN] = snrSpectrogram(sigWithClicks, noiseAudio, nfft,
 nOverlap, 2000, [150 250], []);
            snrContam(k) = 10*log10(rS/rN);
            [rS,rN] = snrSpectrogram(sigCleaned, noiseAudio, nfft, nOverlap,
 2000, [150 250], []);
            snrCleaned(k) = 10*log10(rS/rN);
        case 'spectrogramSlices'
            [rS,rN] = snrSpectrogramSlices(sigWithClicks, noiseAudio, nfft,
 nOverlap, 2000, [150 250], []);
            snrContam(k) = 10*log10(rS/rN);
            [rS,rN] = snrSpectrogramSlices(sigCleaned, noiseAudio, nfft,
 nOverlap, 2000, [150 250], []);
            snrCleaned(k) = 10*log10(rS/rN);
        case 'timeDomain'
            [rS,rN] = snrTimeDomain(sigWithClicks, noiseAudio, [150 250],
 2000);
            snrContam(k) = 10*log10(rS/rN);
            [rS,rN] = snrTimeDomain(sigCleaned, noiseAudio, [150 250], 2000);
            snrCleaned(k) = 10*log10(rS/rN);
        case 'ridge'
            [rS,rN,~,~,~] = snrRidge(sigWithClicks, noiseAudio, nfft,
 nOverlap, 2000, [150 250], []);
            snrContam(k) = 10*log10(rS/rN);
            [rS,rN,~,~,~] = snrRidge(sigCleaned, noiseAudio, nfft, nOverlap,
 2000, [150 250], []);
            snrCleaned(k) = 10*log10(rS/rN);
        case 'synchrosqueeze'
            [rS,rN,~,~,~] = snrSynchrosqueeze(sigWithClicks, noiseAudio, nfft,
 nOverlap, 2000, [150 250], []);
            snrContam(k) = 10*log10(rS/rN);
            [rS,rN,~,~,~] = snrSynchrosqueeze(sigCleaned, noiseAudio, nfft,
 nOverlap, 2000, [150 250], []);
            snrCleaned(k) = 10*log10(rS/rN);
    end
end

fig6 = figure('Position', [250 250 560 400], 'Name', 'Click removal');
x = 1:nM;
bar(x - 0.2, snrContam,  0.35, 'FaceColor', [0.75 0.2
 0.2], 'DisplayName', 'With clicks');
hold on;
bar(x + 0.2, snrCleaned, 0.35, 'FaceColor', [0.2 0.55
 0.2], 'DisplayName', 'After removeClicks');
yline(refSNR, '--k', 'LineWidth', 1.5, 'DisplayName', 'Clean reference');
hold off;
set(gca, 'XTick', x, 'XTickLabel', clickMethods, 'XTickLabelRotation', 15);
ylabel('SNR (dB)');
legend('Location','northeast');
title('Effect of click contamination and removal on each SNR method');
grid on;
```



5. Lurton vs Simple Formula

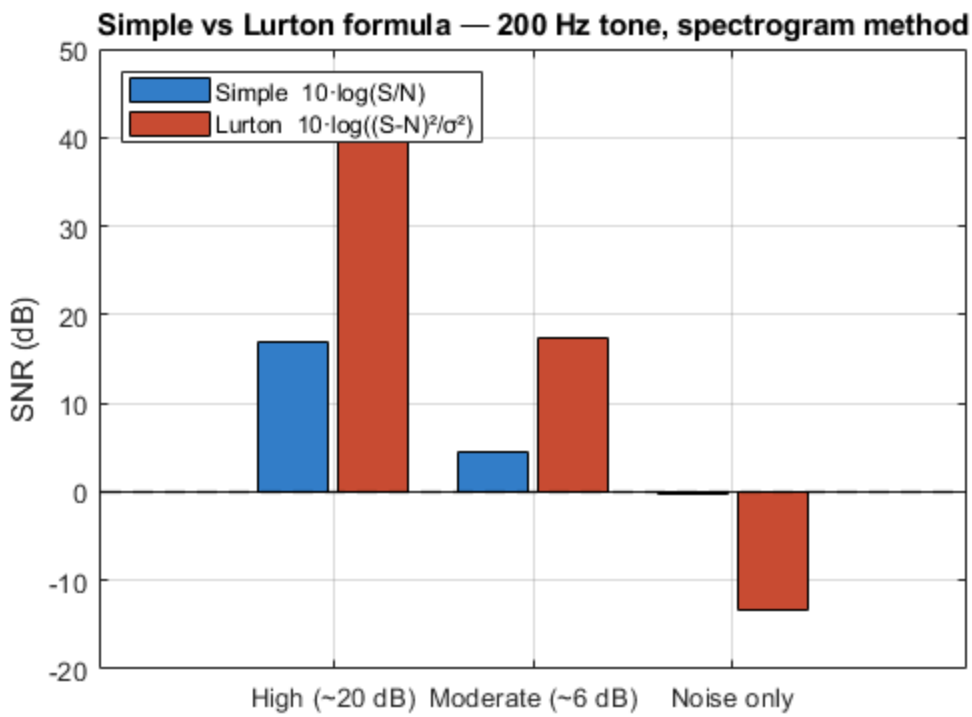
The Lurton (2002) formula, SNR = 10·log10((S#N)²/#²_N), differs from the simple power ratio 10·log10(S/N) in two important ways:

- At **high SNR** it gives much larger values because the squared signal-noise difference amplifies the contrast.
- Near **0 dB SNR** it can give large negative values, clearly flagging detections where no signal is present.
- It depends on the noise **variance** #²_N, making it more sensitive to noise non-stationarity.

```
sigLevels  = {'High (~20 dB)', 'Moderate (~6 dB)', 'Noise only'};
sigRMSVals = [1.0, 0.2, 0.0];
simpleSNRs = nan(1,3);
lurtonSNRs = nan(1,3);

for k = 1:3
    [annotL, cleanupL] = createTestFixture( ...
        'signalRMS', sigRMSVals(k), 'noiseRMS', 0.1, ...
        'toneFreqHz', 200, 'freq', [150 250], 'durationSec', 4);
    simpleSNRs(k) = snrEstimate(annotL, ...
        struct('snrType','spectrogram','showClips',false,'useLurton',false));
    lurtonSNRs(k) = snrEstimate(annotL, ...
        struct('snrType','spectrogram','showClips',false,'useLurton',true));
    cleanupL();
end
```

```
fig7 = figure('Position', [280 280 560 380], 'Name', 'Lurton vs simple
 formula');
x = 1:3;
bar(x - 0.2, simpleSNRs, 0.35, 'FaceColor',[0.2 0.5 0.8],'DisplayName','Simple
  10·log(S/N)');
hold on;
bar(x + 0.2, lurtonSNRs, 0.35, 'FaceColor',[0.8 0.3 0.2],'DisplayName','Lurton
  10·log((S-N)²/#²)');
hold off;
yline(0, '--k', 'LineWidth', 1, 'HandleVisibility', 'off');
set(gca, 'XTick', x, 'XTickLabel', sigLevels);
ylabel('SNR (dB)');
legend('Location','northwest');
title('Simple vs Lurton formula — 200 Hz tone, spectrogram method');
grid on;
```

```
Reading audio metadata for file 1/1: C:\Users\brian_mil\AppData\Local\Temp
\annotSNR_test_20260401_222629_079\2026-04-01_22-26-29.wav
Reading audio metadata for file 1/1: C:\Users\brian_mil\AppData\Local\Temp
\annotSNR_test_20260401_222629_119\2026-04-01_22-26-29.wav
Reading audio metadata for file 1/1: C:\Users\brian_mil\AppData\Local\Temp
\annotSNR_test_20260401_222629_156\2026-04-01_22-26-29.wav
```



6. Method Comparison Grid

This figure compares all six SNR methods (rows) across three signal conditions (columns): high SNR (~20 dB), moderate SNR (~6 dB), and noise only (0 dB). All signals are a 200 Hz tone in wideband noise.

The `timeDomain` row shows bandpass-filtered instantaneous power rather than a spectrogram — signal window in dark green, noise in dark red.

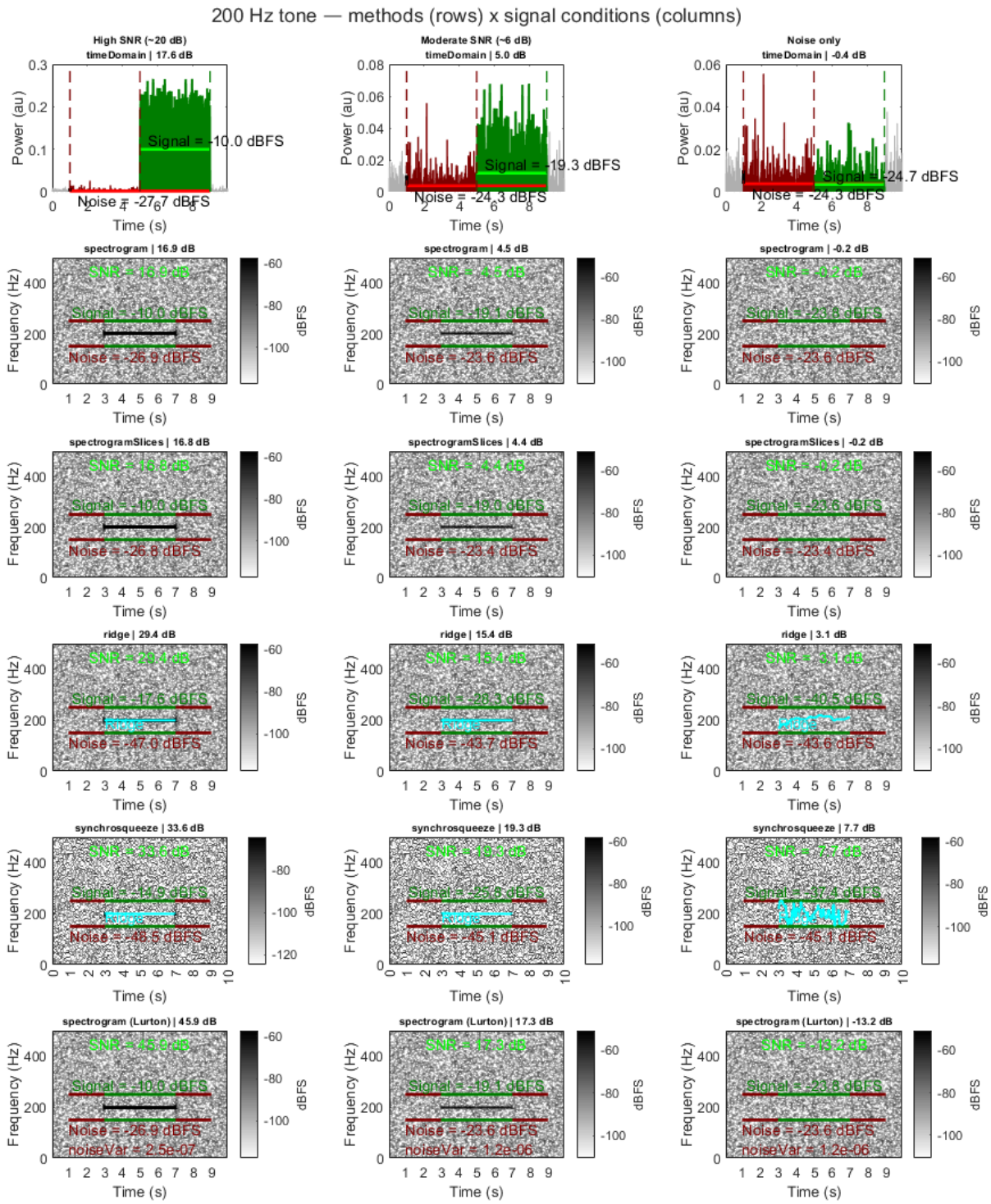
Reading across a row: how each method responds to changing SNR. Reading down a column: how methods differ for the same signal.

```
sigConfigs = {
    'High SNR (~20 dB)',    1.0,  0.1
    'Moderate SNR (~6 dB)', 0.2,  0.1
    'Noise only',           0.0,  0.1
};
nSigs      = size(sigConfigs,1);
gridMethods = {'timeDomain','spectrogram','spectrogramSlices', ...
               'ridge','synchrosqueeze','spectrogram (Lurton)'};
nMethods   = numel(gridMethods);

% Build one fixture per column and keep WAV path for audioread
annots   = cell(nSigs,1);
wavPaths = cell(nSigs,1);
cleanups = cell(nSigs,1);
for s = 1:nSigs
    [annots{s}, cleanups{s}] = createTestFixture( ...
        'signalRMS', sigConfigs{s,2}, 'noiseRMS', sigConfigs{s,3}, ...
        'toneFreqHz', 200, 'freq', [150 250], 'durationSec', 4, ...
        'classification', sigConfigs{s,1});
    d = dir(fullfile(annots{s}.soundFolder, '*.wav'));
    wavPaths{s} = fullfile(annots{s}.soundFolder, d(1).name);
end

fig2 = figure('Position', [130 100 900 1080], 'Name', 'Method comparison
 grid');
tlo2 = tiledlayout(fig2, nMethods, nSigs, ...
    'TileSpacing','compact','Padding','compact');
title(tlo2, '200 Hz tone — methods (rows) x signal conditions (columns)', ...
    'interpreter','none');

nfftG    = 2^nextpow2(floor(4/30/0.75*2000));
nOverlap = floor(nfftG*0.75);
freq     = [150 250];
fs       = 2000;

for k = 1:nMethods
    m        = gridMethods{k};
    isLurton = strcmp(m, 'spectrogram (Lurton)');
    isTD     = strcmp(m, 'timeDomain');
    snrType  = 'spectrogram';
    if ~isLurton, snrType = strtrim(m); end

    for s = 1:nSigs
        ax = nexttile(tlo2);
        annot = annots{s};

        if isTD
            % Load audio with audioread using known fixture timing.
```

```
            % createTestFixture: bufferSec = 0.5*dur + spectroBuffer(1) +
safetyBuffer(2)
            % With durationSec=4: bufferSec=5s, detection at [5 9]s in file.
            % Noise window: immediately before detection (noiseDelay=0), same
duration.
            % Clip for plotTimeDomainPower: noise.t0 - pre to annot.tEnd +
post.
            [fullAudio, fsWav] = audioread(wavPaths{s});
            durSec    = annot.duration;
            bufSec    = 0.5*durSec + 1 + 2;   % 5s for durationSec=4
            detOn     = round(bufSec * fsWav) + 1;
            detOff    = round((bufSec + durSec) * fsWav);
            noiseOff  = detOn - 1;
            noiseOn   = noiseOff - (detOff - detOn);   % same length as
detection

            sigAudio   = fullAudio(detOn:detOff);
            noiseAudio = fullAudio(noiseOn:noiseOff);

            [rmsS, rmsN, nVar] = snrTimeDomain(sigAudio, noiseAudio, freq,
fsWav);
            snr = 10*log10(rmsS/rmsN);

            % Replicate processOne clip timing: noise.t0 - pre to annot.tEnd +
post
            preSec    = sp.pre;
            postSec   = sp.post;
            noiseT0   = annot.t0   - durSec/86400;   % noise window start
datenum
            noiseTEnd = annot.t0;                      % noise window end =
det start
            clipT0    = noiseT0   - preSec/86400;
            clipTEnd  = annot.tEnd + postSec/86400;
            clipOn    = max(1, round((clipT0 - (annot.t0 -
bufSec/86400))*86400*fsWav)+1);
            clipOff   = min(length(fullAudio), round((clipTEnd - (annot.t0 -
bufSec/86400))*86400*fsWav));
            clipAudio = fullAudio(clipOn:clipOff);

            detStruct          = annot;
            detStruct.rmsLevel = 10*log10(rmsS);
            noiseStruct        = struct('t0', noiseT0, 'tEnd', noiseTEnd, ...
                                        'rmsLevel', 10*log10(rmsN));
            plotTimeDomainPower(clipAudio, clipT0, detStruct, noiseStruct, ...
                freq, fsWav, rmsS, rmsN, nVar);
        else
            p = struct('snrType', snrType, 'useLurton', isLurton, ...
                'showClips', true, 'pauseAfterPlot', false, 'spectroParams',
sp);
            snr = snrEstimate(annot, p);
        end

        title(ax, sprintf('%s | %.1f dB', m, snr), ...
            'interpreter','none', 'FontSize', 7);
```

```
        if k == 1
            % Signal condition label above top row
            title(ax, sprintf('%s\n%s | %.1f dB', ...
                sigConfigs{s,1}, m, snr), ...
                'interpreter','none', 'FontSize', 7);
        end
    end
end

for s = 1:nSigs, cleanups{s}(); end
```

```
Reading audio metadata for file 1/1: C:\Users\brian_mil\AppData\Local\Temp
\annotSNR_test_20260401_222629_436\2026-04-01_22-26-29.wav
Reading audio metadata for file 1/1: C:\Users\brian_mil\AppData\Local\Temp
\annotSNR_test_20260401_222629_445\2026-04-01_22-26-29.wav
Reading audio metadata for file 1/1: C:\Users\brian_mil\AppData\Local\Temp
\annotSNR_test_20260401_222629_453\2026-04-01_22-26-29.wav
```



Summary

This gallery has demonstrated:

- Six SNR estimation methods and their sensitivity characteristics

- Method comparison across high, moderate, and zero SNR conditions
- Ridge and synchrosqueeze methods for FM tonal calls (SRW upcall)
- Calibrated acoustic levels (dB re 1 µPa) from instrument metadata
- Click contamination and removal using `removeClicks`
- Simple vs Lurton SNR formula comparison

Function reference: `snrEstimate`, `snrSpectrogram`, `snrSpectrogramSlices`, `snrTimeDomain`, `snrRidge`, `snrSynchrosqueeze`, `removeClicks`, `spectroAnnotationAndNoise`

Test suite: `run('tests/run_tests.m')`

%%%